

ENS 491 – Graduation Project (Design)

Proposal

Project Title:

Implementing Deep Neural Networks (DNNs) Using GPUs

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1. ABSTRACT

Since increasingly more artificial intelligence (AI) is being deployed across industries, the challenge of efficient use of deep neural networks (DNNs) on low resource devices has become an important challenge. IoT devices operating with low power supply and processing resources require low power and low latency solutions in order to provide AI based applications. Traditional cloud based processing is latency and bandwidth intensive, which makes edge computing a preferable choice for real time AI inference. However, executing DNNs on IoT devices require optimized implementations that trade off between computational efficiency and power consumption.

This project explores the acceleration of DNN inference on NVIDIA CUDA, a parallel computing architecture for rapid execution of deep learning workloads on GPUs. The objective is to implement and optimize a selected DNN model, such as MobileNet, EfficientNet, or ResNet, to achieve better performance for edge computing applications. The implementation will focus on matrix operation optimization, parallelism, and memory overhead reduction to maximize execution speed and energy efficiency.

By developing a CUDA based DNN accelerator, this project aims to have a practical solution in terms of AI model deployment in IoT and edge devices. The hope is for improved computational efficiency, reduced latency, and reduced power consumption, making real time deep learning more practical in constrained environments. The result of this project will be a part of the global initiatives of enabling AI driven IoT applications and solving the problems of power efficiency and performance optimization.

2. INTRODUCTION

The combination of AI with the IoT has led to phenomenal expansion in smart applications enabling real time decision making in a lot of areas such as healthcare, self-driving cars, and industrial automation. Running DNNs on IoT devices, however, is riddled with a number of challenges due to their high computational intensity and power usage. Traditional cloud based processing of AI workloads comes with the drawback of introducing large latency, high bandwidth utilization, and even security threats, and hence edge computing becomes an optimal choice for real time applications (Shi et al., 2016). Edge computing provides the possibility of performing AI inference on the IoT devices itself, which saves frequent cloud communication while enhancing responsiveness as well as privacy (Satyanarayanan, 2017). However, executing DNNs on resource-constrained devices remains a problem due to the absence of computational resources, memory, and power constraints.

Several research studies have explored how to optimize DNN inference for edge computing. There are light structures such as MobileNet (Howard et al., 2017), EfficientNet (Tan & Le, 2019), and SqueezeNet (Iandola et al., 2016) that have been used to reduce computation at the expense of maintaining good accuracy. They utilize depthwise separable convolutions, compound scaling, and network compression techniques to promote efficiency. In addition, hardware acceleration techniques such as pruning and quantization have also been applied to compress models and cut down computational requirements (Han et al., 2016). Despite these efforts, solutions available today often come with trade-offs between accuracy, latency, and power consumption. There are also market AI accelerators such as Google's Tensor Processing Units (TPUs) and NVIDIA's Jetson Nano that have specific hardware support for deep learning inference but are marred by closed architecture and astronomical pricing, rendering them inaccessible to generic-purpose IoT applications (Jouppi et al., 2017). Besides, energy-efficient AI deployment is also an upcoming area of research, with new papers exploring new approaches to balance the trade-off between efficiency and accuracy for edge-based deployments (Deng et al., 2020).

While past work has done a lot to optimize DNNs for edge computing, there remain several challenges. Lightweight architectures today are at the cost of accuracy for efficiency and their application is therefore limited in performance-critical applications. Moreover, hardware-centric solutions are less flexible and need special components, thus less suitable for general IoT applications. Although quantization and pruning techniques reduce computational load, they introduce loss of accuracy, which negatively impacts model reliability in safety-critical real time applications (Jacob et al., 2018). Another limitation is the non-use of general purpose GPU architectures for edge AI, although they can potentially offer high performance parallel computation. Against this backdrop, there is a need for more energy-efficient and adaptive deep learning solutions that can leverage existing GPU capabilities to accelerate DNN inference without the need for specialized hardware.

In order to overcome such challenges, the project will architect a GPU based DNN accelerator using NVIDIA CUDA to accelerate deep learning inference for IoT and edge computing applications. By leveraging CUDA's parallel processing capabilities, the project aims to achieve maximum computational efficiency while minimizing energy consumption. Unlike special-purpose AI accelerators, the proposed approach is to optimize general purpose GPUs, enabling a flexible and scalable platform for AI deployment in IoT devices. The aim of this initiative is to enable real time AI in resource limited environments without a performance compromise and without requiring costly hardware upgrades. This project will facilitate the existing research on energy efficient deployment of AI by demonstrating an optimized CUDA version of a selected DNN architecture, providing a real-world and adaptable solution for AI powered IoT applications.

3. PROPOSED SOLUTION AND METHODS

This project is to implement a GPU based deep neural network (DNN) accelerator using NVIDIA CUDA to enable efficient inference on IoT and edge devices. The objective is to optimize deep learning models for real time processing through leveraging GPU parallelism to accelerate the computation and minimize energy consumption. Most recent solutions, such as Google's Tensor Processing Units (TPUs) and NVIDIA's Jetson Nano, provide dedicated hardware for deep learning inference but with the price of high cost, restricted availability, and proprietary architectures (Jouppi et al., 2017). This project, however, proposes a CUDA based software optimization solution that can be applied on general purpose GPUs, offering a scalable and flexible solution for AI deployment on IoT devices.

The scope of the project is to deploy an optimized deep learning inference model using CUDA with optimization emphasis on execution time and energy use. A standard DNN model—such as MobileNet, EfficientNet, or ResNet—will be selected based on its applicability to IoT environments. The implementation will be achieved through matrix computation optimization, minimizing memory overhead, and exploiting GPU parallelism to achieve efficient high-performance inference. But the project is not centered on new neural network architectures or hardware component change. Moreover, although the solution is specially built for IoT and edge computing use cases, it does not concern direct hardware integration with IoT hardware beyond GPU based computation. Instead, the project is centered on software level optimization that can be used for general AI applications across the board.

The complexity of this problem lies in several technical difficulties that involve advanced computational techniques. First, the deployment of DNNs on resource-constrained IoT devices involves several conflicting factors, including low computational resources, energy consumption, memory availability, and real-time processing requirements (Deng et al., 2020). Optimizing neural network inference within these constraints is not straightforward and requires profound knowledge of parallel computing and GPU architecture. Second, modern lightweight DNN architectures achieve efficiency at the cost of reduced accuracy, which limits their usability in performance-critical domains (Howard et al., 2017; Tan & Le, 2019). In addition, quantization and pruning techniques reduce model size but come at the cost of precision loss, making it challenging to maintain inference quality (Han et al., 2016). Third, deep learning inference optimization with GPUs requires solving subtle problems such as memory coalescing, kernel divergence, and optimal thread synchronization—challenges that occur infrequently in regular deep learning model construction.

This project will adopt a formalized approach to resolve these issues. First, the right DNN architecture will be selected depending on performance and efficiency needs. Secondly, the model will be programmed using CUDA, leveraging techniques such as optimized memory

access patterns, shared memory usage, and kernel fusion to speed up execution as well as lower energy consumption (Nvidia, 2020). Performance benchmarking will be done in order to compare optimized GPU based deployment with standard CPU based inference with respect to prominent metrics such as processing time, energy usage, and accuracy maintenance. Finally, analysis of energy efficiency will be done to check if the optimized model can be deployed on IoT devices or not.

Through the development of an efficient CUDA DNN accelerator, this project aims to bridge the gap between deep learning research and real world deployment in IoT. The solution proposed here will also benefit the AI powered edge computing as a whole by providing a high performance, cost effective, and scalable solution compared to application specific AI accelerators. Through this exercise, what is learned will help in optimizing deep learning inference for restricted energy and computation environments.

3.1. Objectives/Tasks

Objective 1: Literature Review of Artificial Intelligence, Machine Learning, and Deep Learning

Intended Outcome: Summarize and understand the concepts of Artificial Intelligence, Machine Learning, and Deep Learning.

Objective 2: Literature Review of Deep Learning

Intended Outcome: Understand fundamental concepts, architectures, and achievements in deep learning, such as ImageNet and AlexNet.

Objective 3: Analysis of Computational Techniques in Deep Learning

Intended Outcome: Investigate various computational techniques, such as parameter scaling, processing needs, and hardware optimizations for deep neural networks (DNNs).

Objective 4: Deploy Deep Learning Models on CPU and GPU

Intended Outcome: Investigate the speed and accuracy difference of PyTorch and TensorFlow deployments on CPU and GPU.

Objective 5: Use EfficientNet, MobileNet, or ResNet with CUDA

Intended Outcome: Design CUDA-based implementations of the selected DNNs to accelerate inference and make optimized execution on GPUs.

Objective 6: Evaluate Edge Computing Performance Metrics

Intended Outcomes: The desired objective is measuring execution performance, energy consumption, memory size, and correctness to analyze if optimized DNNs are deployable on IoT devices.

Objective 7: Learn and Optimize CUDA based Deep Learning

Intended Outcomes: Learn to program CUDA for quick deep learning computations with low-latency and power-conscious edge AI app optimizations.

3.2. Realistic Constraints

To optimize AI inference on edge computing and IoT devices, this project involves the implementation of a deep neural network (DNN) accelerator using NVIDIA CUDA. The development of this project is subject to multiple realistic constraints, including economic, environmental, social, health and safety, manufacturability and sustainability considerations. Therefore, the design, development and implementation of the project is influenced by these constraints.

Economic Constraints

By focusing on CUDA based optimization on general purpose GPUs, this project eliminates the necessity of costly specialized hardware. In addition, software based optimization reduces reliance on custom hardware, maintenance expenses are considerably lower in comparison to dedicated AI accelerators that require specific updates or replacements.

Environmental Constraints

In AI computing energy consumption is an important environmental concern because training and inference of deep learning models require considerable computational resources which result in high electricity usage (Strubell, Ganesh, & McCallum, 2019). The lack of efficiency in traditional DNN inference leads to higher carbon emissions. Thus, optimization is required to minimize energy consumption. With the application of CUDA based optimizations that reduce computation load and power consumption, this project addresses the issue of energy and makes AI more environmentally friendly for edge computing applications.

Social Constraints

In the field of AI, data protection and privacy are very essential, as user collected data can be vulnerable to theft or unauthorized sale. This concern becomes even more significant in IoT environments, where real time inference mostly involves sensitive user data (Shi et al., 2016).

Cloud based AI solutions may be risky due to data being transmitted over public networks, which can expose data to cyber threats. By optimizing AI inference for edge computing, this project enhances data privacy by allowing local processing on devices instead of relying on cloud based services.

Health and Safety Constraints

This project does not have a direct effect on health or safety. Still, deployment of AI in IoT platforms can endanger public safety. Deployment of unoptimized AI inference models to edge computing contexts may result in slowing down real time decision making, which is highly important in cases of autonomous vehicles, healthcare monitoring systems, and industrial automation (Zhou et al., 2019).

Manufacturability Constraints

This project focuses on software based optimization. Therefore, there is no physical manufacturability constraint. However, since the accelerator is designed for IoT applications, it must be deployable on edge devices with limited computational power and memory. In addition, the implementation must be compatible with diverse hardware configurations, including various GPU models and IoT edge devices.

Sustainability Constraints

This project supports social, economic, and environmental sustainability. Optimization of DNN inference for general purpose GPUs, extends the usability of existing hardware, reduces new device production and minimizes electronic waste. Economically, the project's software based approach ensures that AI acceleration is accessible to a broad spectrum of users without necessitating expensive hardwares. Environmentally, optimization reduces energy consumption and decreases overall carbon footprint associated with deep learning computations (Patterson et al., 2021).

3.3. Engineering/Scientific Standards

This project follows a set of accepted engineering and scientific standards to render our CUDA based DNN accelerator trustworthy, accurate, and viable:

IEEE 754 Floating Point Standard

We rely on the IEEE 754 standard for floating point arithmetic that specifies floating point number representation and calculation. Conformity to this standard enables us to provide reproducible and the same output for our DNN computation on both the CPU and the GPU, especially when performing exact matrix multiplications and convolutions. Since floating point arithmetic constitutes the majority of computations in deep learning training and inference,

adherence to IEEE 754 helps prevent undesirable numerical differences and ensures reproducible results (IEEE, 2019).

NVIDIA CUDA Best Practices

Even though NVIDIA CUDA is proprietary, best practices are set and defined in the NVIDIA CUDA C++ Programming Guide and the CUDA Best Practices Guide (NVIDIA, 2025a; NVIDIA, 2025b). The guidelines address critical performance and correctness issues like:

Memory Coalescing: Organizing data in GPU memory to minimize latency and maximize bandwidth.

Shared Memory Utilization: Utilizing on chip shared memory for frequently accessed data to bypass global memory bottlenecks.

Thread and Block Configuration: Having the right kernel launch dimensions (i.e., thread block sizes) to achieve maximum utilization of GPU resources and minimize warp divergence.

Good Synchronization: Reducing unnecessary thread synchronization that will halt parallel performance.

By embracing these best practices, we can maintain our CUDA kernels in top performance, thus minimizing computation time as well as total power consumption—crucial in IoT and edge applications.

Common AI Performance Metrics (Benchmarking)

While not an "official" standard, the AI research community widely refers to accepted benchmarks (e.g., MLPerf) and ubiquitous metrics like latency (inference time per sample), throughput (samples processed per second), and energy per inference. These metrics are what we use to report our GPU accelerated DNNs' improvement in performance. By following traditional benchmarking conventions, we are able to report results both comparably to other state of the art solutions, as well as pertinently to end users.

Compliance with these standards and best practices ensures our project's software element attains proper numerical performance (IEEE 754) and optimal GPU utilization (CUDA Best Practices). It also facilitates reproducibility and simplicity of comparison of our findings with ongoing work in the field of deep learning acceleration.

4. RISK MANAGEMENT

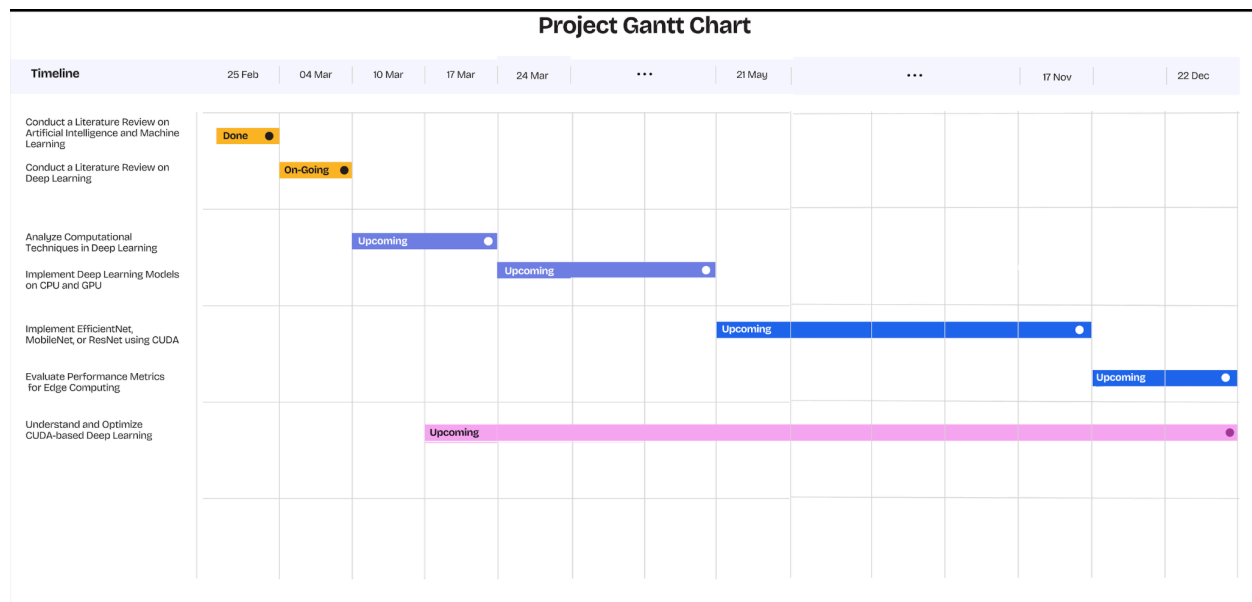
With the complexity of implementing DNNs on resource-restricted devices, managing potential risks becomes part of the project. The project, which explores the acceleration of DNN inference using NVIDIA CUDA for edge computing, encounters challenges in resource restraint, skill scarcity, and time management. The efficient use of AI on limited devices requires access to new research, hardware knowledge, and advanced software tools. If any one of these materials is becoming difficult to access, substitute alternatives such as cloud-based platforms, training materials, and open access repositories will be employed to make progress uninterrupted.

The other challenge is the probable absence of CUDA programming and configuring sophisticated DL frameworks expertise. Since the technologies involved are contemporary, it matters the most that the team must be adequately qualified to deal with technical issues. Where there is limited internal expertise, the project will fill this requirement by organizing specific training sessions, off-site consultations, and utilization of comprehensive online documentation and community resources. In this manner, project momentum is maintained and areas of knowledge shortfall are effectively filled.

5. PROJECT SCHEDULE

Link:

https://www.canva.com/design/DAGg6cef9_c/uLjTFisTR8brzgF-8vYDZQ/edit?utm_content=DAGg6cef9_c&utm_campaign=designshare&utm_medium=link2&utm_source=sharebutton



6. ETHICAL ISSUES

This project does not involve any ethical concerns because it does not collect, process, or store any personal data. However, there are some other ethical concerns to be addressed. For instance, even if not directly working with personal data, the use of popular pre-trained DNN models (such as MobileNet, EfficientNet, or ResNet) may unintentionally spread the existing biases present in the training datasets used for their development, potentially resulting in discriminatory or unfair results in real-world use cases. Additionally, the inherent lack of transparency of deep neural networks also potentially introduces transparency and explainability issues, particularly if such deployments are later to be used within applications where understanding the decisioning process is critical. Finally, although the immediate scope of the work is limited, there is the potential broader implication when using AI with IoT edge-computing capabilities when considering future security exploits or misuse scenarios, which must be projected ahead of time.

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